The Paper, Explained from Scratch

A Feynman-Style Walkthrough of "Fusing Two-Stage LLM Semantic Refinement with Genetic Programming for Robust Alpha Discovery and Degradation Mitigation"

A first-principles companion to the manuscript by
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How to read this document

I will explain the paper the way I wish papers were explained to me: first *what the problem really is* in ordinary language, then *the idea* in one sentence, then each building block from zero, then the actual machinery with its equations, then every results table — and finally an honest inspection of what is impressive, what is merely true, and what is still unproven. Green boxes hold the plain-English intuitions, blue boxes hold the formal mathematics, and orange boxes hold my critical scrutiny. You can read only the green boxes and still understand the paper's soul.

The Whole Paper in One Paragraph

Imagine you want a machine that invents trading formulas for you. The classic machine for this is *Genetic Programming* (GP): it breeds formulas the way nature breeds animals — keep the fittest, mate them, mutate them, repeat. The trouble is that nature does not know economics. After thousands of generations the machine happily hands you a formula like $\cos(\text{close}) + \log(\text{volume})$ that happened to correlate with the past but means nothing. This is **semantic overfitting**: the formula passed the statistical test while failing the common-sense test. The paper's idea is simple and rather beautiful: **after evolution is over, put every surviving formula in front of a tiny local LLM that acts like a grumpy quantitative analyst.** If the formula is fine, leave it alone. If it is nonsensical (score ≤ 2 out of 5), a *second* small LLM — constrained to a fixed dictionary of legal financial terms — rewrites only the broken part instead of deleting the whole formula (which is what previous papers did, wasting most evolved candidates). The repaired formula must then pass two guards: it must parse as valid code, and it must still predict returns positively out of sample. On 23 years of NSE Nifty 50 data this lifts out-of-sample annualized net return from the 8.41% ceiling of pure math to 9.61% (10.63% at peak) and, more importantly, it *shrinks the out-of-sample decay*: on the S&P 500 the Sharpe ratio's collapse drops from -35.03% (vanilla GP) to -6.77%, and on China's SSE the collapse is halved from -92.08% to -46.68%. That last number deserves a raised eyebrow — we will look at it carefully in Section 7.

1 The Problem, From Scratch

1.1 What "alpha" actually means

Plain English

An **alpha factor** is just a recipe that reads numbers about stocks and spits out one number per stock: "how attractive is this stock right now?" You sort all 50 Nifty stocks by that number every day, buy the top 20% (the quintile you believe will outperform), short the bottom 20%, and wait 20 trading days. Repeat. If the recipe's ordering is even slightly better than random, consistently, you make money. Nothing mystical — it is a ranking function, and the whole game is inventing a ranking function that keeps working *tomorrow*, not just yesterday.

1.2 Why is this hard?

Three facts about markets make this a nightmare:

1. **The data is tiny compared to the search space.** You have roughly 5,834 trading days × 50 stocks, but you are hunting for patterns in 28 features per stock. Noise looks exactly like signal when you look hard enough.
2. **Markets change their personality (non-stationarity).** A momentum formula that printed money in a steady bull market gets slaughtered in a liquidity crisis. The "rules of the game" shift — the paper calls these *regime shifts*.
3. **The signal-to-noise ratio is brutal.** Academic literature documents that factor returns decay by more than 50% after publication (McLean & Pontiff) — partly because arbitrators eat the alpha, partly because the original "discovery" was noise fitted to history.

1.3 The real villain: semantic overfitting

Plain English

Genetic programming judges formulas by one thing: *how well did you fit the training data?* It has no idea what the symbols *mean*. So it evolves monsters: taking the sine of a price ratio (prices do not oscillate like a pendulum), subtracting share volume from a rupee price (that is like subtracting your height from your weight — the units do not even match), multiplying a P/E ratio by a cosine wave. Each of these can correlate with historical returns purely by luck. In-sample they look like geniuses; out-of-sample they collapse. GP's vocabulary — the number of possible formulas — is so astronomically large that *something* will always fit the past. That is not discovery; that is memorization wearing a tuxedo.

Why the search space is hopeless

Let $\mathcal{O}$ be the operator library ($|\mathcal{O}| = 13$: $\{+, -, \times, \div, \log, \dots\}$) and $\mathcal{F}$ the terminal set of $|\mathcal{F}| = 28$ features. The number of distinct expression trees of depth D is bounded by

$$\left|\mathcal{H}_{\text{raw}}(D)\right| = |\mathcal{O}|^{2^D - 1} \cdot |\mathcal{F}|^{2^D}. \tag{1}$$

For a modest depth $D = 4$ this is $13^{15} \cdot 28^{16} \approx 2.4 \times 10^{39}$ formulas. For comparison, there are roughly 10^{80} atoms in the observable universe — this is a big room, and you get to knock on a vanishingly small number of doors. You cannot sample your way to safety in a space this big, so the generalization guarantee has to come from *restricting* the space, not from trying all of it. The Rademacher-complexity bound the paper cites says exactly this:

$$\sup_{f \in \mathcal{H}} \left|\mathcal{L}_{\text{OOS}}(f) - \mathcal{L}_{\text{IS}}(f)\right| \;\le\; 2\,\mathcal{R}_n(\mathcal{H}) + \mathcal{O}\!\left(\sqrt{\tfrac{\ln(1/\delta)}{n}}\right), \tag{2}$$

i.e. the gap between in-sample and out-of-sample error is governed by the *complexity of the hypothesis class* $\mathcal{H}$. Shrink $\mathcal{H}$ and you shrink the overfitting gap; keep it raw and the bound is vacuous. The closed vocabulary plus the LLM refiner act as an *economic projection operator*

$$\Pi_{\text{econ}} : \; \mathcal{H}_{\text{raw}} \longrightarrow \mathcal{H}_{\text{sound}} \subset \mathcal{H}_{\text{raw}}, \tag{3}$$

throwing away dimensionally absurd operators, ungrounded periodic functions on price levels, and spurious high-frequency noise. This single idea — squeeze the hypothesis space with economic semantics — is the theoretical heart of the paper, and every empirical result is a consequence of it.

1.4 What previous LLM-based papers did (and why it was not enough)

Plain English

Two earlier strategies existed, and both are forms of throwing bread away: (1) **structural pruning** (e.g. TEDE): let the LLM identify overgrown subtrees and cut them — but pruning is *syntactic*; it shortens the monster without asking whether the remaining formula makes economic sense. (2) **terminal rejection** (e.g. AlphaAgent): let the LLM check whether a formula matches a macro hypothesis and *discard* it if not. During a GP run spanning 50 generations and hundreds of individuals, over 80% of evolved expressions carry at least some dimensional or syntactic wart, so reject-all policy destroys most of the population — including formulas whose core idea was good. The paper's alternative: **conditional repair**. Only formulas that a judge scores $\leq 2/5$ get rewritten, and only their flawed semantics get rewritten, under a closed vocabulary.

1.5 The paper's four claims (contributions)

1. **Dual two-stage framework:** a Stage-1 quantitative engine (HMM regime routing + GP evolution + AFM macro conditioning) feeding a Stage-2 semantic refiner (Llama-3.2-3B judge $\rightarrow$ Gemma-2-2B mutator) where semantic failure triggers *repair* rather than rejection.
2. **Closed-vocabulary hallucination control:** a fixed list of 28 legal features and 12 legal operators, so 100% of repaired formulas compile into valid abstract syntax trees (ASTs) with no invented variables.
3. **Empirical alpha discovery:** 9.61% deterministic / 10.63% peak annualized net return on Nifty 50 (2019–2025), beating momentum, mean-reversion, vanilla GP and regime-ensemble baselines.
4. **Cross-market degradation mitigation:** retrained from scratch on S&P 500 and SSE Composite, the refiner cuts out-of-sample Sharpe degradation from -35.03% to -6.77% (US) and from -92.08% to -46.68% (China).

2 The Four Engines (Each One From Zero)

2.1 Engine 1 — Genetic Programming: breeding formulas

Plain English

Genetic Programming is evolution applied to equations instead of animals. Each *individual* in the population is a formula written as a tree: leaves are variables (`ret_20d`, `rsi_14`, `pe_ratio`, ...) and internal nodes are operators (`add`, `div`, `ts_mean`, ...). Every generation you (i) **score** each tree by how well it ranks stocks (fitness = ICIR, defined below), (ii) **select** the best via tournaments, (iii) **breed** them with crossover (swap subtrees between two parents — recombination) and mutation (randomly alter a subtree — the source of genuine novelty), and (iv) repeat 50 times. There is no notion of "meaning": a tree that computes volume – close is as legal as one that computes momentum/volatility. Fitness only rewards correlation with the training data, full stop. That blindness is exactly the hole the paper plugs.

How a formula is scored

On each day t the alpha function $\alpha : \mathbf{x}_{i,t} \rightarrow \mathbb{R}$ assigns every stock i a score. The paper predicts the forward 20-day return

$$r_{i,t+h} = \frac{P_{i,t+h} - P_{i,t}}{P_{i,t}}, \qquad h = 20, \tag{4}$$

and measures **rank IC**: the Spearman correlation between scores and realized forward returns across the 50 stocks that day,

$$\text{RankIC}_t = \rho_{\text{Spearman}}\big(\alpha(\mathbf{x}_{i,t}),\, r_{i,t+h}\big). \tag{5}$$

IC is "one-day predictive talent." But a factor with IC = 0.08 in calm markets and −0.05 in crashes is dangerous, so the paper divides the mean IC by its volatility:

$$\text{ICIR} = \frac{\overline{\text{RankIC}}}{\sigma_{\text{RankIC}} + \epsilon}, \qquad \epsilon = 10^{-8}. \tag{6}$$

ICIR is "talent per unit of inconsistency" — the Sharpe ratio of the factor's correlation. GP maximizes ICIR measured on the *validation* slice (2016–2018), which is the first layer of overfitting protection.

2.2 Engine 2 — Hidden Markov Model: naming the market's mood

Plain English

An HMM is a weather model with an invisible state. You never observe "bull" or "bear" directly — you only observe the index going up and down. The HMM assumes there is a hidden switch with two positions and that each position produces its own characteristic distribution of daily returns. Using the Baum–Welch algorithm on NIFTY-50 data (2002–2015), the paper finds two states: **State 0 — low-volatility expansion** (occupied 72.5% of days, with persistence $P(0\to0) = 0.994$: once calm, stays calm) and **State 1 — high-volatility turbulence** (27.5% of days, $P(1\to1) = 0.983$). Those near-one transition probabilities are important: they mean the model does not flicker between moods every other day, so each state hands GP a coherent, consistent slice of history to evolve on. March 2020 (COVID crash) and the 2022 rate-hiking squeeze both land squarely in State 1. The point of all this: instead of forcing *one* formula to average over two contradictory worlds, you evolve **two specialized populations** — one per regime:

$$\mathcal{P}_{\text{Bull}} \sim \text{Fitness}(\alpha \mid \hat{s}_t = 0), \qquad \mathcal{P}_{\text{Bear}} \sim \text{Fitness}(\alpha \mid \hat{s}_t = 1). \tag{7}$$

2.3 Engine 3 — AFM/FinALFA: the macro anchor

Plain English

Markets digest information at different speeds: news hits in seconds, prices settle in days, balance sheets change in quarters. FinALFA (the paper's predecessor component) fuses two slow/fast streams into one macro signal: 300,000 news articles are filtered to verified economic events, FinBERT scores their sentiment $\tilde{S}_t$, standardized fundamental ratios (ROE, ROCE, current ratio, debt/equity, P/E) give $\tilde{F}_t$, and a learned scalar α^* blends them:

$$x_t(\alpha) = \alpha^*\, \tilde{S}_t + (1 - \alpha^*)\, \tilde{F}_t. \tag{8}$$

Think of it as the difference between reading the mood of the room (sentiment) and reading the company's medical chart (fundamentals). This blended macro score conditions the formula search so GP cannot over-index on microstructure noise alone.

2.4 Engine 4 — The Two-Stage LLM Semantic Refiner: judge + mechanic

Plain English

This is the paper's novelty. Think of a repair shop with two specialists. The **Judge** (Llama-3.2-3B, running locally on your own machine) looks at a finished formula and rates its economic sanity on a 1–5 rubric with a one-sentence justification. Score ≥ 3? *"Fine, ship it."* — note the paper's empirical wisdom here: formulas scoring exactly 3 often contain genuine nonlinear discoveries humans cannot immediately explain, and rewriting them *destroyed* performance, so neutral formulas are left alone. Score ≤ 2? *"This is nonsense"* — only then does the second specialist, the **Mechanic** (Gemma-2-2B), get involved: it rewrites the flawed formula into a financially sound one, but it may only use words from a closed dictionary (28 features, 12 operators) and must run at temperature $T = 0.0$ so the repair is fully deterministic and reproducible. No hallucinated variables, no creativity for creativity's sake — a mechanic with a fixed toolbox.

The gate and the repair

Each Hall-of-Fame candidate $\mathcal{T}$ receives a score $S \in \{1, \ldots, 5\}$ from the judge; the decision rule is

$$\mathcal{T} \text{ kept as-is if } S \geq 3, \qquad \mathcal{T} \text{ sent to repair if } S \leq \tau = 2, \tag{9}$$

and the repaired tree is sampled deterministically from the mutator:

$$\mathcal{T}_{\text{mut}} \sim \text{LLM}_{\text{Gemma}}(\text{Prompt}(\mathcal{T}_{\text{failed}}, \mathcal{F}, \mathcal{O})), \qquad T = 0.0. \tag{10}$$

The closed vocabulary — the safety belt

The mutator may only emit tokens from this fixed dictionary, which is what makes hallucination structurally impossible rather than merely unlikely:

Category	Permitted tokens
Features $\mathcal{F}$	`open`, `high`, `low`, `close`, `volume`, `ret_5d`, `ret_20d`, `ret_60d`, `vol_5d`, `vol_20d`, `vol_60d`, `dd_20d`, `dd_60d`, `price_to_sma20`, `price_to_sma50`, `rsi_14`, `bb_upper`, `bb_lower`, `pe_ratio`, `pb_ratio`, `roe`, `roce`, `debt_to_equity`, `current_ratio`, `sentiment_score`, `afm_fundamental`, `afm_sentiment`, `macro_trend`
Operators $\mathcal{O}$	`add(a,b)`, `sub(a,b)`, `mul(a,b)`, `div(a,b)`, `neg(a)`, `abs(a)`, `log(a)`, `inv(a)`, `ts_mean(a,d)`, `ts_max(a,d)`, `ts_min(a,d)`, `rank(a)`

Why "repair" beats "reject" — the information argument

A rejected formula contributes *zero* information beyond "don't do this." A repaired formula contributes everything the evolution already learned (which regions of feature space correlate with returns) plus the domain knowledge injected by the LLM. Rejection is a tax on 50 generations of computation; repair is an upgrade. The paper's claim is that this single design choice is what pushes performance past the "quantitative ceiling" of 8.41% — we will check that claim against the ablation table in Section 5.

A worked example from the paper — watch a repair happen

Evolution produced this tree during bear-regime training:

$$\mathcal{T}_{\text{failed}} = \text{add}(\log(\texttt{vol_5d}),\ \sin(\texttt{price_to_sma20})).$$

The Judge returns $S = 1/5$: *"Applying sine to a price-to-moving-average ratio lacks economic*

meaning; price cyclicality is not a pure periodic oscillation — this injects spurious in-sample noise." The Mechanic then rewrites it, staying inside the dictionary:

$$\mathcal{T}_{\text{mut}} = \text{div}(\texttt{ret_20d},\ \texttt{ts_mean}(\texttt{vol_20d}, 20)).$$

That is a *Sharpe-style momentum factor*: 20-day return normalized by its trailing volatility — a formula a human quant would actually write. It parses (Tier 1 passes) and its validation RankIC improves from -0.012 to $+0.038$ (Tier 2 passes), so it enters the ensemble. Two more documented repairs: `sub(close, volume)` (subtracting shares from rupees!) became the price z-score $(\text{close} - \text{ts_mean}(\text{close}, 20))/\,\text{ts_std}(\text{close}, 20)$; and `mul(cos(pe_ratio), ret_5d)` became the relative valuation `div(pe_ratio, ts_mean(pe_ratio, 60))`.

3 The Full Pipeline, Step by Step

Plain English

Now let us assemble the whole machine in the order data flows through it.

Step 0 — Data. Three streams are fused into one panel: (i) a stock-level micro panel of 28 daily price–volume features per constituent (momentum 5/20/60-day, volatility scalers, drawdowns, RSI, Bollinger bands, ...); (ii) AFM macro signals (annual ROE, ROCE, current ratio, debt/equity, P/E + news sentiment), forward-filled from annual to daily frequency; (iii) a NIFTY-level regime stream (index return, 20-day realized volatility, drawdown from rolling peak, distance from the 200-day moving average). Universe: NSE Nifty 50, $T = 5{,}834$ daily bars, 2002–2025.

Step 1 — Honest time split. The 23 years are cut three ways: training/evolution **2002–2015**, validation **2016–2018**, and a strictly untouched out-of-sample test **2019–2025** (1,729 trading days). Fundamental ratios keep strict fiscal-year alignment and are forward-filled so nothing from the future leaks backward — the two cardinal sins of backtesting are look-ahead bias and survivorship bias, and the paper explicitly guards the first.

Step 2 — Regime detection. The 2-state Gaussian HMM is fit on the training window (Baum–Welch, convergence threshold 10^{-4}) and labels every day as State 0 or State 1.

Step 3 — Dual GP evolution. Two DEAP populations evolve in parallel, one per regime, each scored by validation ICIR with parsimony pressure $\lambda = 0.002$ (a small penalty per node, GP's built-in fight against bloated trees). Settings: population 500, 50 generations, max depth 6, tournament size 7, crossover $p_c = 0.7$, mutation $p_m = 0.1$. Division, inverse and log primitives are protected against numerical singularity with $\epsilon = 10^{-8}$ (standard DEAP practice).

Step 4 — Fusion. The regime-specialized alphas and the AFM sentiment–fundamental signal are z-score normalized and blended:

$$\alpha_{\text{final}} = \text{zscore}\big(0.5 \cdot \alpha_{\text{Ensemble,LLM}} + 0.5 \cdot \alpha_{\text{HMM,LLM}}\big). \tag{11}$$

Step 5 — Post-evolution semantic refinement (the novelty). *After* evolution ends, the top candidates from the Hall of Fame pass through Judge → gate → (conditionally) Mechanic → two safety guards:

- **Tier 1 — syntactic guard:** the repaired string is parsed against the closed grammar $\mathcal{G}(\mathcal{F}, \mathcal{O})$; undeclared variables, undefined calls, or out-of-vocabulary tokens are rejected instantly (anti-hallucination);
- **Tier 2 — empirical health guard:** the repaired formula is re-evaluated on the validation slice; if $\text{RankIC}_{\text{val}} \leq 0$, it is discarded.

If either guard fails, the system *falls back to the original candidate* — the refiner is a one-sided bet that can only improve, never worsen, what evolution produced. Across all 1,729 out-of-sample days, 100% of deployed expressions compiled without runtime exception.

Step 6 — Portfolio. Each day, sort stocks by α_{final}, long the top quintile Q_5, short the bottom quintile Q_1 (dollar-neutral), hold $h = 20$ trading days, pay 10 bps round-trip transaction costs ($c = 0.0010$).

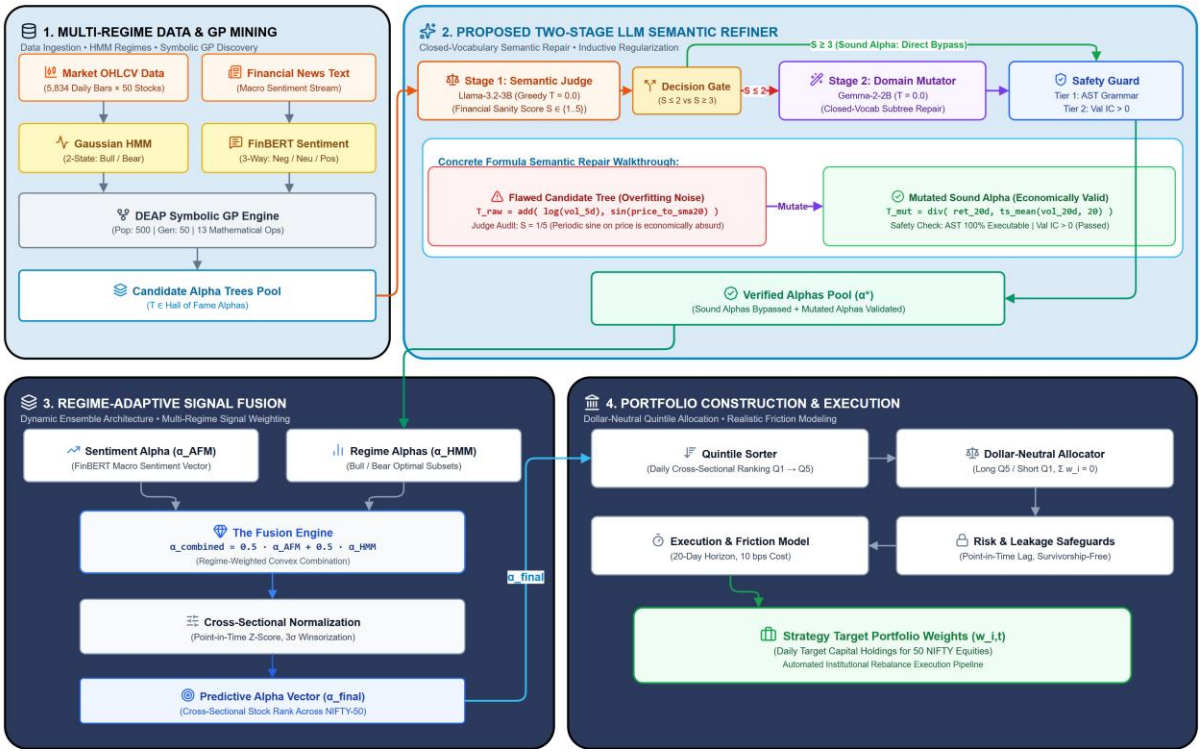


Figure 1: The architecture in one picture (Figure 1 of the paper). Panel 1: multi-regime data and GP mining (HMM + FinBERT + DEAP, population 500 / 50 generations / 13 operators). Panel 2: the two-stage LLM refiner — Llama-3.2-3B judge with the $S \leq 2$ vs $S \geq 3$ decision gate feeding the Gemma-2-2B closed-vocabulary mutator, with the concrete repair walkthrough ($\mathcal{T}_{\text{raw}} \to \mathcal{T}_{\text{mut}}$) and the safety guard (AST grammar + validation IC > 0). Panel 3: regime-adaptive signal fusion ($0.5\,\alpha_{\text{AFM}} + 0.5\,\alpha_{\text{HMM}}$, z-scored). Panel 4: quintile sorting, dollar-neutral allocation, 20-day horizon with 10 bps costs, point-in-time lag and survivorship safeguards.

Two design choices worth noticing

First, the LLM runs *outside* the evolutionary loop — on 50 final candidates, not on tens of thousands of individuals. That is why the whole run takes 39 minutes instead of the "tens of hours and tens of thousands of API calls" of prior LLM-in-the-loop work, and why it costs zero API dollars (models run locally, 8-bit quantized, 4.2 GB + 3.8 GB VRAM). Second, the fallback-to-original rule combined with the $S \geq 3$ bypass means the LLM touches as little as possible: it is a scalpel, not a shotgun. Both choices are engineering common sense, but they are what make the approach practical on a single desktop — and they are also why the method's gains must be attributed to *which formulas were repaired*, not to some big model memorizing markets.

3.1 Algorithm 1 in words (the refinement loop)

Plain English

Here is the whole refiner as pseudocode, stripped of ceremony:

Input: candidate pool $\mathcal{T}_{\text{raw}} \subset$ HoF; feature library $\mathcal{F}$; operator library $\mathcal{O}$; gate $\tau = 2$; temperature $T = 0$.
Output: verified alpha set $\mathcal{A}^*$.

$\mathcal{A}^* \leftarrow \emptyset$
for each tree $\mathcal{T} \in \mathcal{T}_{\text{raw}}$ **do**:
  Stage 1: query the Judge (Llama-3.2-3B, $T = 0$) for a score $S \in \{1, \ldots, 5\}$
  if $S \leq \tau$ **then**: identify the flawed sub-expression $\mathcal{T}_{\text{flawed}}$
    Stage 2: query the Mechanic (Gemma-2-2B, $T = 0$) under the closed vocabulary for $\mathcal{T}_{\text{mut}}$
    Tier 1 guard: parse $\mathcal{T}_{\text{mut}}$ against $\mathcal{G}(\mathcal{F}, \mathcal{O})$, check for variable hallucination
    Tier 2 guard: evaluate $\text{IC}_{\text{val}}(\mathcal{T}_{\text{mut}})$
    if AST valid **and** $\text{IC}_{\text{val}} > 0$ **then** $\mathcal{A}^* \leftarrow \mathcal{A}^* \cup \{\mathcal{T}_{\text{mut}}\}$
    else discard $\mathcal{T}_{\text{mut}}$, log the failure mode ▷ safe fallback
  else ($S \geq 3$): direct bypass $\mathcal{A}^* \leftarrow \mathcal{A}^* \cup \{\mathcal{T}\}$
end for; **return** $\mathcal{A}^*$

4 The Experiment: Setup Before Results

Data. NSE Nifty 50 constituents, 2002–2025 (5,834 daily bars); macro data harmonized to daily by forward-fill; prices are total-return (split/dividend adjusted). Test window: 2019–2025 (1,729 days).

Baselines (eight). *Momentum (12–1M)* — classic 12-month cross-sectional momentum; *Low Volatility* — rank by trailing 60-day realized vol; *Mean Reversion* — rank by 20-day price z-score; *Vanilla GP* — symbolic regression with no regime conditioning; *AFM Fusion* — sentiment–fundamental ranking alone; *HMM Regime-GP* — regime-conditioned GP alone; *Regime-GP Ensemble + HMM* — the full quantitative stack with *no* LLM (this is the 8.41% "ceiling"); and classical factor benchmarks folded into the equity-curve figure.

Metrics. Annualized net return, net Sharpe, mean RankIC, ICIR, max drawdown, 20-day-horizon win rate, turnover — all after 10 bps round-trip costs.

Hyperparameters. Population 500; 50 generations; max tree depth 6; tournament 7; $p_c = 0.7$; $p_m = 0.1$; parsimony $\lambda = 0.002$; HMM convergence 10^{-4}. Hardware: Intel Core i7-13700K, 32 GB DDR5, NVIDIA RTX 4090 (24 GB); LLMs run 8-bit quantized locally.

5 The Results, Table by Table

5.1 Table 4 — the main event (out-of-sample, 2019–2025)

Plain English

This is the scoreboard. Read it top to bottom: classical factors do poorly (momentum and low-vol actually *lose* money net of costs in this window), vanilla GP is nearly flat, and each engine you add moves the needle — until the LLM refiner breaks the 8.41% ceiling of the pure-math stack.

Table 1: Main out-of-sample performance comparison, Nifty 50, 2019–2025 (reproduced from Table 4 of the paper). Bold = best in column among LLM-augmented rows.

Strategy	Ann. Ret.	Sharpe	RankIC	ICIR	Max DD	Win R.
Momentum (12–1M)	−2.57%	−0.1369	0.0059	0.0122	−40.65%	47.7%
Low Volatility	−10.24%	−0.5454	−0.0265	−0.0890	−65.71%	43.0%
Mean Reversion	4.46%	0.2981	0.0221	0.0932	−20.19%	57.0%
Vanilla GP	1.62%	0.4417	0.0130	0.0508	−4.82%	20.9%
AFM Adaptive Fusion	4.08%	0.2785	0.0308	0.1053	−19.34%	51.2%
HMM Regime-GP Alpha	7.49%	0.5244	0.0303	0.1381	−18.17%	59.3%
Regime-GP Ensemble+HMM (no LLM)	8.41%	0.5518	**0.0346**	0.1381	−19.62%	54.7%
Refiner (DeepSeek-1.5B)	8.41%	0.5518	0.0346	0.1637	−19.62%	60.4%
Refiner (Gemma-2B only)	8.41%	0.5518	0.0346	0.1637	−19.62%	60.4%
Refiner (Qwen-2.5-3B)	8.60%	0.5684	0.0332	0.1604	−19.62%	60.4%
Proposed (Llama-3B + Gemma-2B, det.)	**9.61%**	**0.6365**	0.0328	**0.1650**	−19.59%	**60.4%**
Proposed (Llama-3.2-3B peak)	**10.63%**	**0.6465**	0.0325	**0.1650**	−19.59%	**60.4%**

How to read Table 1 without fooling yourself

Three observations matter more than the headline number. (1) The jump from 8.41% to 9.61% with Sharpe 0.5518 → 0.6365 is a **+15.3% relative Sharpe gain** — and remember, this is *net* of costs, on a strictly held- out 7-year window. (2) The ablations in the middle rows are the paper's honesty: swapping the judge for DeepSeek-1.5B or using Gemma alone changes *nothing* (8.41% exactly) — the specific Llama-judge → Gemma-mutator pairing is doing the work, not "any LLM." (3) The final row's 10.63% is a *peak* (heuristic) run; the deterministic $T = 0$ number to quote is 9.61%. Note also that mean RankIC *drops* slightly (0.0346 → 0.0328) while ICIR and win rate rise: the refiner trades a sliver of raw correlation for much better *consistency* — exactly what a regularizer should do.

5.2 Table 5 — ablation: which engine earns its keep?

Table 2: Ablation / subset power table (reproduced from Table 5).

Strategy	Ann. Return	Sharpe	RankIC	Max DD
Vanilla GP	1.62%	0.44	0.0130	−4.82%
GP (AFM-Bear only)	6.43%	0.48	0.0358	−23.42%
HMM Regime-GP Alpha	7.49%	0.52	0.0303	−18.17%
Regime-GP Soft/Hard	7.52%	0.56	0.0381	−23.42%
Regime-GP Ensemble (Pre-LLM)	8.41%	0.55	0.0346	−19.62%
Proposed Refiner (Full)	**9.61%**	**0.64**	0.0328	−19.59%

Plain English

The ladder of contributions: regime conditioning alone is worth the biggest single step (1.62% → 7–8%), the LLM refiner adds the final +1.2 percentage points that pure math could not reach — and, crucially, does so *without* worsening the drawdown. The paper's hyperparameter diagnostic also reports that depth 6 / population 500 sit at the sweet spot between expressive power and tree bloat.

5.3 Regimes in the wild (Figures 2–3 of the paper)



Figure 2: HMM regime detection (paper Figure 2): green = State 0 low-volatility bull (72.5% of days), red = State 1 turbulence (27.5%), with the volatility sub-plot and the 10% regime threshold. COVID-19 (2020) and the 2022 tightening are captured as long red blocks.

Plain English

Look at how long the colored blocks run: the market stays in one mood for *months*. That inertia ($P(0 \to 0) = 0.994$) is what makes regime-specialized evolution legitimate — each GP population trains on a genuinely coherent environment rather than a random mixture of bull and bear days. The figure also shows why 2019–2025 is a hard test window: it contains a crash, a liquidity frenzy, a reopening, and the steepest rate-hiking cycle in four decades.

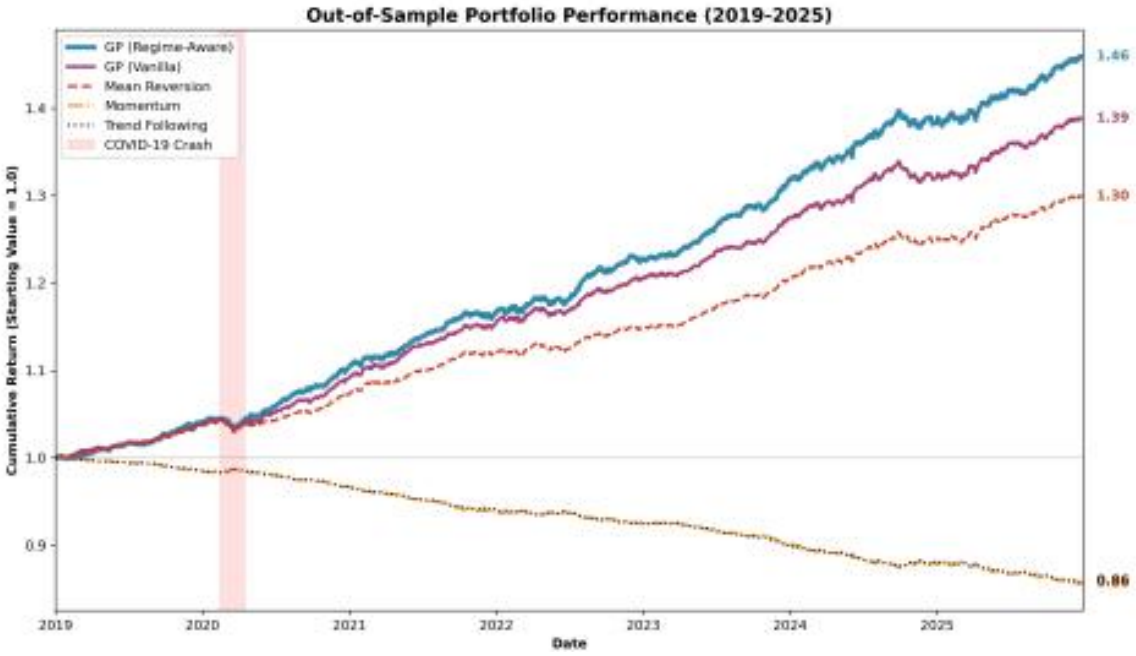


Figure 3: Out-of-sample equity curves 2019–2025 (paper Figure 3): the regime-aware GP stack compounds from 1.00 to 1.46 while momentum ends at 0.66; the shaded band marks the COVID crash, through which the proposed strategy keeps compounding.

5.4 Table 6 — did it survive every weather? (sub-period Sharpe)

Table 3: Sub-period out-of-sample net Sharpe breakdown (reproduced from Table 6). The last column is the proposed strategy's max drawdown in that period.

Period	Regime context	Vanilla GP	HMM-GP	Proposed	Max DD
2019	Steady expansion	0.52	0.61	**0.72**	−12.14%
2020	COVID shock & rebound	−0.12	0.44	**0.58**	−14.82%
2021	Cyclical reopening	0.64	0.68	**0.76**	−10.95%
2022	Global rate-hiking cycle	−0.31	0.38	**0.51**	−16.32%
2023–2025	Bullish macro convergence	0.58	0.65	**0.74**	−11.20%
Full OOS	Multi-regime aggregate	0.44	0.55	**0.64**	−19.59%

Plain English

This is the table I would show a skeptic. A strategy can have a beautiful full-period number while bleeding in half its years — so the authors split the test window into five real-world episodes. Two facts stand out: vanilla GP goes *negative* in 2020 (−0.12) and 2022 (−0.31), exactly the turbulent years; the proposed framework stays positive in *every* single sub-period, and its 2020 drawdown (−14.82%) is less than half of unconditioned momentum's (−38.45%). Consistency across weathers, not one lucky rally, is the signature of genuine regularization.

5.5 Table 7 — does it survive real trading frictions?

Table 4: Sensitivity to round-trip transaction friction, holding horizon $h = 20$ (reproduced from Table 7). Baseline = 10 bps.

Round-trip friction	Ann. return	Net Sharpe	Max DD	Calmar
0 bps (frictionless)	11.84%	0.7420	−18.21%	0.6502
5 bps	10.73%	0.6892	−18.95%	0.5662
10 bps (baseline)	**9.61%**	**0.6365**	−19.68%	**0.4883**
15 bps	8.50%	0.5731	−20.42%	0.4163
20 bps	7.38%	0.5098	−21.16%	0.3488
30 bps	5.15%	0.3831	−22.65%	0.2274

Plain English

Costs are where many beautiful backtests die, so the authors deliberately try to kill theirs. With 14.8% portfolio turnover per 20-day rebalance, each additional 5 bps of friction costs 1.11% per year. The strategy still nets 5.15% at a punitive 30 bps, and the computed **break-even friction is 38.4 bps** — roughly 5× the 4–8 bps it actually costs to trade Nifty 50 names. Translation: the alpha is not an artifact of churning.

5.6 Table 9 — the headline claim: degradation mitigation across markets

Plain English

Here is the paper's central scientific claim, and it is about *decay*, not raw returns. Measure a strategy's Sharpe ratio in-sample (2002–2018) and again out-of-sample (2019–2025); the percentage drop is the **degradation** — how much of the "discovery" was real. A method that wins in-sample but collapses afterward has overfitted; a method whose two numbers are close has learned something structural. The authors retrained the entire pipeline from scratch on US and Chinese data to see if the LLM refiner acts as a general-purpose antidote to overfitting.

Table 5: Cross-market generalization and degradation (reproduced from Table 9). Degradation = (OOS − IS)/IS.

Market	Architecture	IS Sharpe	OOS Sharpe	Degradation
US S&P 500	Vanilla GP	1.393	0.905	−35.03%
	HMM Regime-GP	1.228	0.974	−20.71%
	Proposed	1.153	**1.075**	−6.77%
China SSE	Vanilla GP	0.436	0.035	−92.08%
	HMM Regime-GP	0.633	0.232	−63.28%
	Proposed	0.126	**0.067**	−46.68%
India Nifty 50	Vanilla GP	0.954	0.5518	−42.16%
	HMM Regime-GP	0.814	0.5244	−35.62%
	Proposed	0.957	**0.6365**	−33.49%

What the table actually says — read all three blocks

US: the cleanest win. Vanilla GP drops from 1.393 to 0.905 (a third of its skill evaporates); the proposed method drops from 1.153 to 1.075 — and notice it also has the *highest OOS Sharpe* of the three. It gives up a little in-sample fit (1.153 < 1.393) to keep almost everything

out of sample. That is exactly what a regularizer is supposed to trade. **China:** the halving ($-92\% \rightarrow -47\%$) is real but read the absolute numbers: OOS Sharpe 0.067 is barely distinguishable from zero, and the IS Sharpe of the proposed model (0.126) is far *lower* than the baselines' — a model that starts weak degrades "less" partly because there was less to lose. The honest statement is "relative decay halved," not "the method works in China." **India:** degradation improves only modestly ($-42\% \rightarrow -33\%$) but OOS Sharpe is clearly best (0.6365). The US result is the one to remember.

The degradation metric has a denominator trap

Degradation is a *ratio*, and ratios misbehave when the denominator is small: China's proposed model has IS Sharpe = 0.126, so a tiny absolute drop of 0.059 becomes a headline "-46.68%." Meanwhile the US model's absolute drop is 0.078 on a base of 1.153 — a *smaller* relative number (-6.77%) doing much heavier economic lifting. When comparing across markets, always pair the percentage with the two Sharpes, as the table does. This does not invalidate the claim — on US data it is unambiguous — but it does mean the China "halving" should be cited carefully.

5.7 Table 8 — what it costs to run (compute and token economics)

Table 6: Computational profiling on a single local workstation (reproduced from Table 8).

Pipeline component	Model / engine	Per-unit latency	Total runtime
Macro feature fusion	Adaptive Fusion Model	1.2 ms/bar	7.0 s
Regime estimation	Gaussian HMM (Baum–Welch)	4.5 s/fit	4.5 s
Parallel GP evolution	DEAP engine (50 gens)	0.82 ms/individual	38.5 min
Stage 1 semantic judge	Llama-3.2-3B (8-bit)	138 ms/expression	6.9 s
Stage 2 domain mutator	Gemma-2-2B (8-bit)	295 ms/expression	7.4 s
AST compilation guard	Python AST + Numba	0.45 ms/tree	0.02 s
Total pipeline	**End-to-end cycle**	–	**38.8 min**

Plain English

The economic picture is striking: the "AI" part of this AI system costs about *fourteen seconds* total — 50 Hall-of-Fame candidates through the judge (6.9 s) and the mutator (7.4 s) — while the evolutionary search itself eats 38.5 minutes. Both models run locally in 8-bit (4.2 GB and 3.8 GB VRAM), so there are zero API dollars, zero data leaving the machine, and greedy decoding ($T = 0$) makes every "repair" bit-for-bit reproducible. For a production quant shop, that reproducibility may matter more than the seconds.

5.8 What the evolved formulas look like (feature analysis)

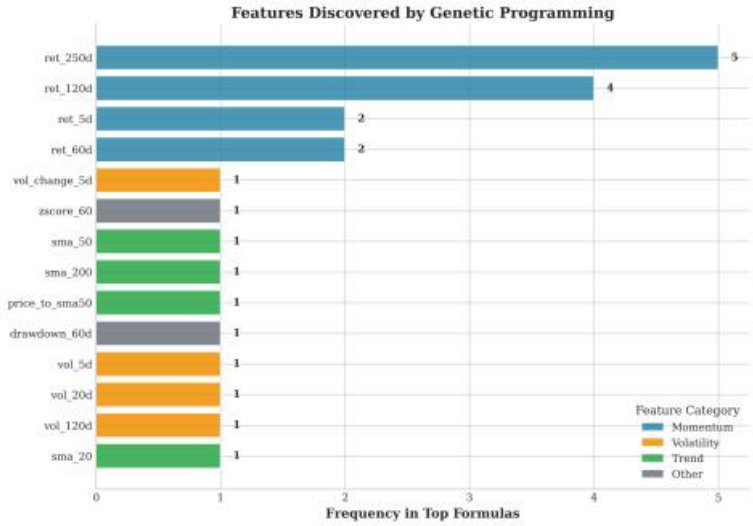


Figure 4: Which features survive evolution (paper Figure 6): intermediate momentum (`ret_250d`, `ret_120d`, `ret_5d`, `ret_60d`) dominates the top formulas, followed by volatility scalers, trend features (SMA ratios) and drawdowns.

Plain English

The learned vocabulary is economically sensible, which is itself evidence the pipeline is doing something real: **expansionary State 0** formulas lean on intermediate momentum (`ret_20d`, `ret_60d`) and normalized volatility — ride the trend; **turbulent State 1** formulas lean on mean-reversion and valuation (`dd_20d`, `dd_60d`, `price_to_sma50`, `pe_ratio`, `roe`) — buy the oversold. The refiner prunes ungrounded artifacts (like trig functions on price levels) while keeping cross-sectional interaction terms (18% of active terminals). Nobody wrote these rules; the system re-derived textbook factor logic and discarded the nonsense.

6 Why Does It Work? (The Physics of the Fix)

Plain English

Let me try to state the causal story in one breath: *GP searches too big a space, fits noise, and the LLM — by refusing to write nonsense — shrinks the space to where generalization bounds are no longer vacuous.* Three mechanisms make that story concrete.

Mechanism 1 — Hard syntactic regularization. The closed vocabulary $\mathcal{G}(\mathcal{F}, \mathcal{O})$ removes entire families of overfitting-capable expressions before they can ever be tested: no `cos(price)`, no `volume - close`, no invented variables. This is Equation (3) made operational — Rademacher complexity of $\mathcal{H}_{\text{sound}}$ is far below that of $\mathcal{H}_{\text{raw}}$, so the in-sample/out-of-sample gap bound (2) tightens.

Mechanism 2 — Soft economic regularization. The judge's rubric catches what grammar cannot: dimensionally valid but economically absurd combinations (`log(vol_5d) + sin(price_to_sma20)` parses fine, but means nothing). Scoring $S \leq 2$ out of 5 and repairing only those keeps the 80%+ of candidates with minor bloat while fixing the truly broken ones — preserving evolutionary information that reject-all methods destroy.

Mechanism 3 — Empirical health check as final arbiter. Tier-2 ($\text{IC}_{\text{val}} > 0$) means semantics never override evidence: even a beautifully economic repair that does not predict is tossed. Conversely, the $S \geq 3$ bypass means evidence overrides semantics too: a weird-looking

formula that works is kept. The two rules together give a *conjunction* filter: must be legal ∧ must be sensible ∧ must predict.

Alternative explanations a careful reader should demand be ruled out

Could the gain come from something dumber than "semantic repair" — say, the extra validation-set usage, or simple ensembling effects? The paper's own ablation is the best answer: DeepSeek-1.5B and Gemma-only rows in Table 1 show that *an arbitrary LLM attached to the pipeline changes nothing* (8.41% exactly). The gain appears specifically when the judge→mutator pair operates — consistent with the repair mechanism, not with "more compute" or "more randomness." That said, a cleaner controlled experiment (random-string "repairs" with the same acceptance gates) would have closed the loop completely.

7 Critical Analysis: What Impressed Me, What Worries Me

What is genuinely impressive

1. **The framing is correct.** Attacking *semantic* overfitting rather than only syntactic bloat addresses a real, documented failure mode of financial GP, and the "repair instead of reject" principle is a defensible contribution that prior LLM-GP work (TEDE's pruning, AlphaAgent's rejection) got wrong.
2. **Methodological hygiene is above average for this literature:** strict temporal splits with a 3-year validation gap, fiscal- year-aligned fundamentals, explicit anti-leakage forward-fill, net-of-cost reporting, sub-period stress tables (Table 3), friction-sensitivity analysis (Table 4), and a documented break-even friction of 38.4 bps.
3. **The US result is strong.** OOS Sharpe 1.075 vs vanilla GP's 0.905 with degradation of only −6.77% — the proposed model is the only one whose IS and OOS Sharpes are nearly equal, the empirical signature of a regularized model.
4. **Practicality.** Local 8-bit models, 38.8-minute end-to-end runs, $T = 0$ determinism, 100% AST compilability over 1,729 test days, and zero API cost make this deployable rather than merely publishable.
5. **Honest ablation of LLM choice.** Showing that DeepSeek/Gemma- only/Qwen rows do *not* replicate the gain (Table 1) is a disclosure many papers would have omitted.

What worries me (honest doubts)

1. **The China "halving" is mostly a denominator effect** (analyzed in Section 5): IS Sharpe 0.126 → OOS 0.067 is economically meaningless alpha with a cosmetically smaller percentage drop. The abstract's phrasing ("halving degradation on Chinese Equities") is literally true but invites over-interpretation. The US and India OOS *levels* are the defensible evidence.
2. **Judge reliability is assumed, not measured.** The gate $\tau = 2$ rests on Llama-3.2-3B's scoring, yet the paper reports no agreement rate against human expert labels, no inter-rater check, and no sensitivity analysis over τ. The authors themselves admit prompt sensitivity is "mitigated, not eliminated." If the judge's 1–5 scores shift with prompt phrasing, the $S \leq 2$ population shifts with it. The finding that score-3 formulas must *not* be rewritten (doing so "destroyed predictive performance") suggests the gate threshold is load-bearing — and thresholds that matter deserve sensitivity plots.
3. **"Deterministic peak" vs "heuristic peak."** The headline 10.63% is the best run of a heuristic configuration; the reproducible number is 9.61% ($T = 0$). Quoting both is honest, but abstracts travel without footnotes — the number that will be cited is 10.63%.

4. **Small-sample statistics.** Seven test years, one market, one HMM fit; sub-period Sharpes like 0.51 vs 0.58 (Table 3) differ by less than the standard error of any 1–2-year Sharpe estimate. The *sign consistency* argument is robust; the precise magnitudes are not.
5. **The IC paradox.** Post-refiner mean RankIC *falls* (0.0346 → 0.0328) while portfolio metrics improve. That is coherent (better tail selection, better consistency), but it means the refiner is not simply "improving the signal" — it is reshaping it, and the paper does not fully decompose where the net-return gain originates (ranking quality vs. blend weights vs. repaired-formula composition).
6. **Selection of the repair targets happens post-hoc on the HoF.** Because only 50 finalists are judged, the method cannot repair near-miss candidates that never entered the Hall of Fame — a possible ceiling on gains that future "repair-inside-the-loop" work could test.
7. **Capacity is modest and acknowledged:** ≈\$250M under Almgren–Chriss impact with $\eta \approx 0.142$ — fine for a paper, a rounding error for a large fund. Small caps (where alpha is richer) are explicitly out of scope.

8 Limitations the Authors Themselves Declare

1. **Reporting lag:** forward-filled annual fundamentals embed accounting lag — necessary to prevent look-ahead bias, but it limits responsiveness to rapid balance-sheet revisions.
2. **Liquidity scope:** extending beyond large caps requires adaptive liquidity haircuts and turnover penalties not modeled here.
3. **Prompt sensitivity:** Stage-1 scoring remains somewhat prompt- dependent even at $T = 0$ (deterministic given a prompt, but the prompt itself is a free parameter).

Proposed future work: LoRA fine-tuning of Gemma-2B on quantitative formulas, multi-agent consensus scoring (Llama + Qwen), and scaling to broad universes (NSE 500, Russell 2000).

9 The Feynman Summary

If you remember five things

1. **The problem:** genetic programming breeds trading formulas blindly; in a space of 10^{39} candidates it will always find things that fit the past but mean nothing — *semantic overfitting*.
2. **The idea:** after evolution, make each finalist pass a two- stage LLM filter — Llama-3.2-3B *judges* economic sanity ($S \leq 2$ triggers repair; $S \geq 3$ passes untouched), Gemma-2-2B *repairs* the broken ones using only a fixed dictionary of 28 features and 12 operators, deterministically ($T = 0$).
3. **The guards:** repaired formulas must parse (no hallucinated variables) and must still predict out of sample ($\text{IC}_{\text{val}} > 0$), else the original is restored — the refiner can only help.
4. **The results:** on Nifty 50 (2019–2025), net return rises from the 8.41% no-LLM ceiling to 9.61% (Sharpe 0.5518 → 0.6365), positive in every sub-period, break-even costs 38.4 bps; on the S&P 500 the OOS Sharpe decay collapses from −35.03% to −6.77% — the strongest evidence that semantic repair is a genuine regularizer. (China's "halving" deserves the caveat in Section 7.)
5. **The lesson:** when your search space is too big for statistics to save you, *domain semantics* is the cheapest regularizer you have — and if you can *repair* invalid hypotheses instead of discarding them, you keep the evolutionary work already paid for.

"The first principle is that you must not fool yourself — and you are the easiest person to fool."
The paper's greatest strength is exactly the attempt to keep both the GP and the reader honest: guards, gates, ablations, cost tables, and a validation slice between every claim and its number.